Machine Learning

Information Systems & Business Analytics 2403 : Machine Learning with Python

Project - Weather Forecasting

By (Group 9):
Tanya Jain
Mehak Agrawal
Kalyani Shinde
Padma Rishitha Pusapati

Introduction

Weather forecasting is an essential aspect of modern society, influencing agriculture, transportation, disaster management, and public safety. Reliable weather predictions help in mitigating risks associated with extreme conditions such as storms, floods, and heatwaves, ultimately saving lives and resources.

Traditional forecasting techniques rely on numerical weather prediction (NWP) models that solve complex mathematical equations based on atmospheric physics. While effective, these methods require significant computational power and are often limited by uncertainties arising from incomplete data, chaotic weather patterns, and resolution constraints.

Machine learning (ML) offers a promising alternative by leveraging vast amounts of historical meteorological data to identify hidden patterns and correlations that conventional models might overlook. ML-based approaches can improve prediction accuracy, reduce computational costs, and enable faster decision-making by automating weather classification tasks.

This study aims to explore ML algorithms for classifying weather conditions into four categories: Sunny, Cloudy, Rainy, and Snowy. By evaluating different models and optimizing their performance, this project demonstrates how ML can be integrated into modern weather forecasting systems for enhanced accuracy and efficiency. Weather forecasting is an essential aspect of modern society, influencing agriculture, transportation, disaster management, and public safety. Traditional forecasting techniques rely on numerical simulations, which can be computationally intensive and sometimes inaccurate due to inherent limitations.

This project explores machine learning (ML) as a potential alternative to improve forecasting efficiency and accuracy. By analyzing meteorological data, our model aims to classify weather conditions (Sunny, Cloudy, Rainy, Snowy) using advanced classification algorithms.

Dataset Overview & Preprocessing

Dataset Characteristics

- Total Samples**:** 13,200
- Total Features**:** 11 (7 numerical, 4 categorical)
- Weather Classes**:** 4 (Sunny, Cloudy, Rainy, Snowy)

Numerical Features: Temperature, Humidity, Wind Speed, Precipitation, Atmospheric Pressure, UV Index, Visibility

Categorical Features: Cloud Cover, Season, Location, Weather Type (Target)

Performed EDA to understand numerical feature distributions, pair plot for numerical variables, predictive power score analysis distribution plots for numerical features, weather type distribution by numerical features and categorical feature distributions.

Data Preprocessing

- **Outlier Detection:** Applied the IQR method, identifying significant outliers in Temperature (92), Wind Speed (404), Atmospheric Pressure (927), and Visibility (383).
- **Feature Engineering:** StandardScaler used for numerical features; OneHotEncoder for categorical features.
- **Class Balancing:** SMOTE (Synthetic Minority Oversampling Technique) applied to address data imbalance.
- **Data Splitting:** Training (72%), Validation (8%), Testing (20%).

Machine Learning Models and Implementation

To classify weather conditions into **Sunny, Cloudy, Rainy, and Snowy**, we experimented with multiple classification models ranging from traditional algorithms to advanced ensemble techniques.

Algorithms Used:

- Logistic Regression
- SVM
- KNN
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost
- LightGBM *(Best performing model)*
- CatBoost

Model Implementation:

All models were implemented using various machine learning frameworks, utilizing specialized libraries for efficient training and evaluation. The dataset was systematically divided into:

- **Training Set:** 72% of data used for model learning.
- **Validation Set:** 8% of data for hyperparameter tuning.
- **Testing Set:** 20% of data for evaluating final performance.

The implementation followed these steps:

1. **Data Preprocessing** (Scaling, One Hot Encoding, Label Encoding)
2. **Feature Selection** (Identifying key meteorological variables)
3. **Training Each Model** on the processed training dataset
4. **Evaluating Models** using accuracy, precision, recall, and F1-score
5. **Hyperparameter Tuning** for improving performance

Hyperparameter Tuning:

To optimize model performance, hyperparameters were tuned using **GridSearchCV** (Exhaustive search over parameter combinations)

Key Hyperparameters Tuned:

Model	Hyperparameters Tuned
Logistic Regression	C (Regularization parameter), Solver (saga)
SVC	Kernel (RBF), C (Regularization parameter)
KNN	Number of neighbors (k), Weights
Random Forest	Number of Estimators (Trees), Max Depth
Gradient Boosting	Learning Rate, Number of Estimators
XGBoost	Learning Rate, Number of Estimators, Max Depth
LightGBM	Learning Rate, Number of Estimators, Max Depth
CatBoost	Depth, Iterations, Learning Rate
Decision Tree	Max Depth, Min Samples Split

Model Training and Evaluation

Each model was trained on the training dataset, validated using the validation set, and tested on the final test dataset.

Model Performance Summary:

Model	Training Accuracy	Test Accuracy	Test F1 Score
LGBMClassifier	0.966961	0.909848	0.909916
DecisionTreeClassifier	0.965383	0.906818	0.906861
GradientBoostingClassifier	0.95181	0.904167	0.904257
XGBClassifier	0.964331	0.90303	0.903106
RandomForestClassifier	0.956229	0.898485	0.899283
CatBoostClassifier	0.940657	0.898106	0.89822
SVC	0.951073	0.894697	0.895129
KNeighborsClassifier	0.923822	0.874621	0.875087
LogisticRegression	0.875947	0.855682	0.855704

Final Model Selection

After evaluating all models, the **LightGBM Classifier** emerged as the best-performing model. It achieved the highest **test accuracy (90.98%)** and **F1-score (90.99%)**, indicating its strong ability to generalize unseen data.

Reasons for Selecting LightGBM:

1. **Highest Test Accuracy & F1-Score**
   - LightGBM outperformed other models, showing superior prediction capabilities.
2. **Balanced Performance Across Metrics**
   - While some models had close performance (e.g., Decision Tree and XGBoost), LightGBM consistently maintained the best balance between accuracy, recall, and precision.
3. **Efficient Training and Scalability**
   - LightGBM is optimized for large datasets and faster training, making it ideal for real-world deployment.
4. **Avoids Overfitting**
   - Despite a high training accuracy (96.69%), its test accuracy remained strong, suggesting effective generalization without significant overfitting.